



Parktown Boys' High School

District: 9

Geography Department

Grade 12 Map Work Assessment

Examiner: Ms A Clayton
Moderator: Mr A Meintjes

Time: 1 Hour
Total: 60 Marks
Date: March 2026

Question	Content	Marks	Assessor Marks	Moderator Marks
1	Multiple choice questions	10		
2	Map calculations	20		
3	Map and photo interpretation	20		
4	Geographical information system	10		
	Total	60		
EDUCATOR'S SIGNATURE:				
DATE:				

RESOURCE MATERIAL

1. An extract from topographic map 2531BC MALELANE
2. Orthophoto map 2531 BC 16 RIVERSIDE

INSTRUCTIONS AND INFORMATION

1. The question paper consists of 10 pages and a map insert.
2. Answer ALL the questions.
3. Answer all questions in the answer booklet (folio paper).
4. You must hand the topographic map and the orthophoto map to the invigilator at the end of this test session.
5. Show ALL calculations and use the formulae provided, where applicable. Marks will be allocated for these.
6. Indicate the unit of measurement in the final answer of calculations, e.g. 10 km; 2,1 cm, West of True North.
7. You may use a non-programmable calculator.
8. You may use a magnifying glass.
9. The area demarcated in RED on the topographic map represents the area covered by the orthophoto map or the mapped area.
10. The following English terms and their Afrikaans translations are shown on the topographic map:

ENGLISH	AFRIKAANS
Diggings	Uitgrawings
Caravan park	Karavaanpark
Sewage works	Rioolwerke
Golf course	Gholfbaan
Wetland	Vleiland

BACKGROUND INFORMATION ON MALELANE



Coordinates: 25°29'S ; 31°31'E

Malelane is a gateway to the Kruger National Park, Mozambique and Eswatini (formerly Swaziland). This charming farming town in Mpumalanga is situated on the N4 national highway about halfway between the capital city of Mpumalanga, Mbombela (formerly Nelspruit), and the capital city of Mozambique, Maputo.

Malelane sits on the intersection of three roads that lead to many of southern Africa's favourite tourist destinations. Driving into Malelane from Nelspruit, you can see the hilly outcrop directly ahead, with the Mozambique border post nestled at its base. Turn left and the promise of a few glorious days in the Kruger National Park beckons you. Turn right and the lush, tropical landscape of Eswatini lies on the horizon.

Malelane, features a hot, subtropical climate with around 547mm of annual rainfall. Temperatures are high, typical of the Lowveld, with, for example, January averaging around 30°C and winter days remaining warm.

[Source: <https://hamiltonslodge.co.za/history-of-malelane/>]

QUESTION 1 – MULTIPLE CHOICE (10 marks)

- 1.1 The questions below are based on the 1: 50 000 topographical map 2531BC MALELANE as well as the orthophoto map 2531 BC 16 RIVERSIDE. Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number on your answer booklet.
- 1.1 Malelane is located in the ... province.
- A. Limpopo
 - B. North West
 - C. Free State
 - D. Mpumalanga
- 1.2 The grid reference number located to the east of the Malelane map 2531BC is:
- A. 2531BD
 - B. 3125BD
 - C. 2531BB
 - D. 3125BD
- 1.3 The actual distance between the **spot height 308** in block **E1** and the **spot height 329** in block **E4** on the orthophoto map is ... metres.
- A. 13.1
 - B. 0.13
 - C. 1300
 - D. 2600
- 1.4 The scale of the topographic map:
- A. 1 cm represents 500 m
 - B. 1 cm represents 500 cm
 - C. 1 m represents 100 m
 - D. 1 cm represents 100 cm
- 1.5 The coordinates of recreation ground in block **D1** on the topographical map is ...
- A. 25°29'44" E 31°35'18" S
 - B. 25°28'18" S 31°30'18" E
 - C. 25°28'42" S 31°30'18" E
 - D. 25°30'18" E 31°31'25" S

1.6 The scale of 1 : 50 000 shows a ... area and ... detail as it is a smaller scale than 1 : 10 000.

- (i) larger
- (ii) smaller
- (iii) less
- (iv) more

- A. (i) and (iii)
- B. (i) and (iv)
- C. (ii) and (iii)
- D. (ii) and (iv)

1.7 The contour interval on the orthophoto map is:

- A. 5m
- B. 10m
- C. 15m
- D. 20m

1.8 Malelane in block **E1** has a/an ... street pattern.

- A. grid
- B. planned irregular
- C. radial
- D. unplanned irregular

1.9 The major primary activity visible in the mapped area is ...

- A. fishing
- B. mining
- C. forestry
- D. crop farming

1.10 The slope formed between **7** and **8** on the orthophoto map is a/an ... slope.

- A. concave
- B. terraced
- C. convex
- D. even

(10 x 1) (10)

[10]

QUESTION 2 – CALCULATIONS (20 marks)

- 2.1 Give the true bearing from the railway station in Malelane in block **E1** to spot height 522 in block **E4**.
(1x2) (2)

- 2.2 Using your answer in QUESTION 2.1, use the information on the topographical map and information below to calculate the **current** magnetic bearing from the railway station in Malelane in block **E1** to spot height 522 in block **E4**. Show ALL calculations.

Mean magnetic declination 18 28' West of True North (January 2014).

Difference in years: **(1)**

Mean annual change 5' Westwards.

Total change: **(1)**

Present magnetic declination: **(2)**

Magnetic Bearing: **(1)**

$$\text{Current MB} = \text{TB} + \text{Current MD}$$

(1x5) (5)

- 2.3. Calculate the area covered by orthophoto on the topographical map. Give the answer in km². Show ALL calculations.

$$\text{Area} = \text{length} \times \text{breadth}$$

(4 x 1) (4)

- 2.4 What is the difference in height between the contour line at **7** and the spot height at **8** on the orthophoto map?
(1x1) (1)

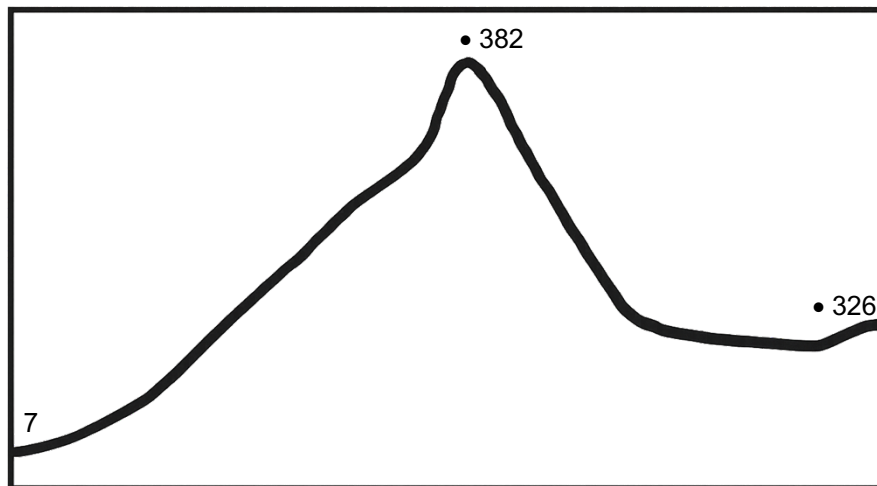
- 2.5 Use the answer to QUESTION 2.4 to calculate the gradient from the contour line at **7** to spot height **8**. Show ALL calculations

Use the map distance of 4,5 cm.

$$\text{Average Gradient} = \frac{\text{vertical interval (VI)}}{\text{horizontal equivalent (HE)}}$$

(3x1) (3)

- 2.6 Refer to the cross-section from the orthophoto map from 7 in block **C4** to spot height 382 in block **E5** below and answer the questions that follow.



Vertical scale: 1cm = 20m

- 2.6.1. If you stand at spot height 382, will you be able to see the person at spot height 326? Give a reason for your answer.

(1x2) (2)

- 2.6.2. Calculate the vertical exaggeration of the cross-section shown above. Show ALL calculations.

$$\text{Vertical Exaggeration} = \frac{\text{vertical scale}}{\text{horizontal scale}}$$

(1x3) (3)

[20]

QUESTION 3 – MAP AND PHOTO INTERPRETATION (20 marks)

3.1 Refer to the Umgwena river in the topographical map.

3.1.1. In which direction is the river flowing to?

(1x1) (1)

3.1.2. Provide evidence from the map to support your answer in QUESTION 3.1.1.

(1x2) (2)

3.2 3.2.1. A wind that blows at night from spot height 405 in block **B3** in a westerly direction to the Umgwenya River, is known as a/an... wind.

- A. berg
- B. anabatic
- C. katabatic
- D. fohn

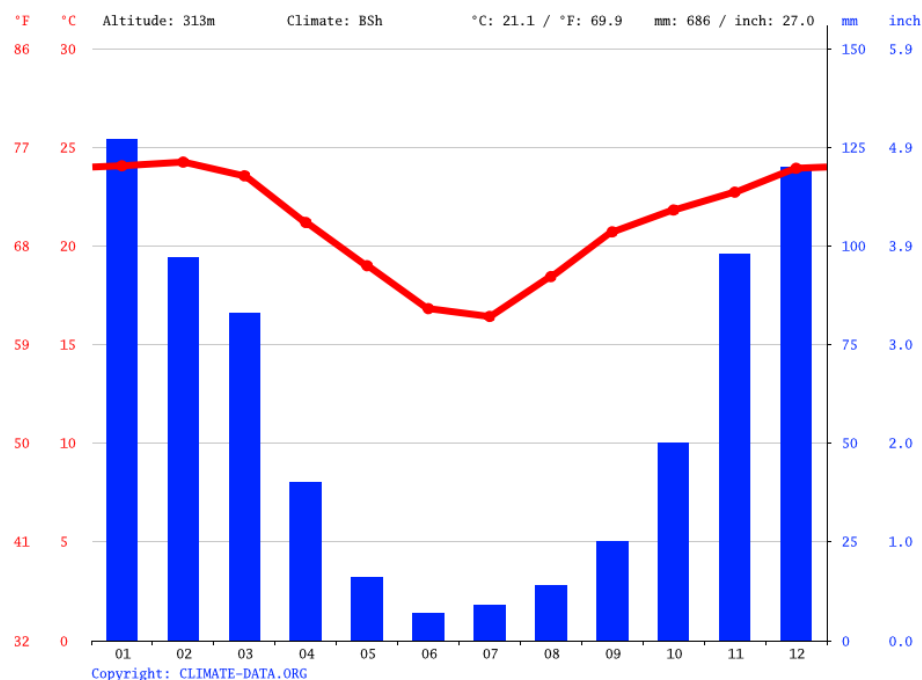
(1x1) (1)

3.2.2. What negative environmental impact could the wind identified in QUESTION 3.2.1 have on the Leopard Creek Golf Estate in block **B2**?

(1x2) (2)

3.3 Refer to the annual climate graph for Malelane below and answer the questions that follow.

Note: Temperature is depicted by the line graph and rainfall is depicted by the bar graph.



[Source: <https://en.climate-data.org/africa/south-africa/malelane-26824/>]

3.3.1. Malelane does not receive rainfall all year long.

In which season does Malelane receive most of its rainfall?

(1x1) (1)

3.3.2. Using the topographical map, give ONE piece of evidence that shows that the area does not receive rainfall all year round.

(1x2) (2)

3.4 In winter, pollution levels are increased due to the inversion layer cause by the subsidence of the Kalahari anticyclone over the region.

3.4.1. Using the topographical map, identify the main cause of the air pollution in the area.

(1x1) (1)

3.4.2. Explain why the inversion layer causes pollution to be worse in winter in the area.

(2x2) (4)

3.5 Refer to block E1 and E2 on the orthophoto map.

3.5.1. Which time of the day (morning/midday/afternoon) was the photograph taken?

(1x1) (1)

3.5.2. Give a reason for your answer to QUESTION 3.5.1.

(1x2) (2)

3.6 Refer to the orthophoto map. Is the feature at 9 a golf course or a dam?

(1x1) (1)

3.7 Give a reason for the choice of location for the golf course.

(1x2) (2)

[20]

QUESTION 4 – GEOGRAPHIC INFORMATION SYSTEMS (10 marks)

4.1 Refer to area covered by blocks B3 and C3 on the topographical map.

4.1.1. Define the concept data layering.

(1x2) (2)

4.1.2. Name one data layer found in block **D5** and block **E5**.

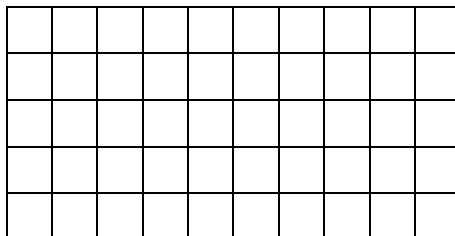
(1x1) (1)

4.2 Refer to diagram **A** and gridblocks 1 and 2 below.

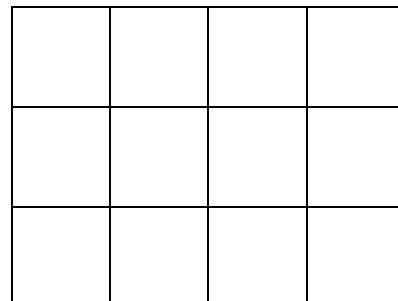
Diagram A



Gridblock 1



Gridblock 2



4.2.1. Explain the concept spatial resolution?

(1x2) (2)

4.2.2. The spatial resolution of image in Diagram A is **high**.

Match the gridblocks at **1** or gridblocks at **2** with the resolution represented by diagram. Support your answer with a reason

(1 +2) (3)

4.3 Malelane is frequently impacted by tropical cyclones in summer. Suggest **ONE** method on how the Malelane municipality can use Remote Sensing to raise awareness to the public on an approaching tropical cyclone.

(1x2) (2)

[10]